

Continuous Estimation of Monkey Hand Position using Discrete Classification and State-based Trajectory Estimation

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Abstract—We present a lightweight two-stage decoder that reconstructs arm kinematics from intracortical spikes recorded while a non-human primate performed an eight-direction centre-out reaching task ($N = 98$ units, 100 training and 82 test trials per direction). Spike trains are first binned in 10 ms windows and variance-stabilised with the Anscombe transform. A time-resolved linear discriminant analysis (LDA) classifier, updated every 20 ms from 320 ms to 560 ms from initial recording, predicts the intended reach direction with 98.86 % accuracy at the final latency. The classified direction selects a trajectory regressor that projects cumulative counts onto 20 principal components and applies ordinary least-squares weights; this yields a hand-position root-mean-square error of 8.22 cm and a run-time of 22.54s, resulting in a runtime-weighted RMSE (Task Score) of 9.65. Gaussian-process and support-vector regressors offered marginal accuracy gains but exceeded real-time computational budgets. Ablation studies confirmed that performance is robust to bin-width variation, principal-component truncation and 20 % channel loss. The resulting pipeline runs in under 30 s of training time per fold and meets the latency requirements of closed-loop brain-machine interfaces, providing a practical balance between accuracy, speed and implementation simplicity.

I. INTRODUCTION

Intracortical brain-machine interfaces (BMIs) transform neuronal population activity into control commands for external devices, offering a route to restore motor function after paralysis [1]. While multi-electrode arrays now provide millisecond-resolution recordings from hundreds of cortical units, turning these high-dimensional, stochastic spike trains into smooth, continuous kinematics remains difficult. Decoder performance can be degraded by spike-count variability, non-stationary tuning and the conflicting demands of fast categorical decisions versus precise continuous estimates [2].

Recent work has shown that splitting the problem into a *discrete* movement-class decoder followed by a *state-based* trajectory estimator improves accuracy and computational efficiency [3]. Building on these insights, we present a modular decoding framework that combines (i) a time-resolved linear discriminant analysis (LDA) classifier for rapid identification of the intended reach angle and (ii) a principal-component, linear-regression stage that continuously estimates the two-dimensional hand trajectory. The classifier operates on incrementally summed spike counts using a step size of 20 ms, capturing the evolving population response without incurring the latency associated with fixed-window methods. For each classified direction, a dedicated regression model - trained in the reduced principal-component space - maps cumulative firing activity onto Cartesian hand coordinates.

The remainder of the paper is organised as follows. Section II details the experimental data set, preprocessing routine and the training of the time-resolved classifiers and regressors. Section III reports the decoding accuracy, trajectory error and ablation analyses that evaluate the influence of bin width and dimensionality on performance. Finally, Section IV discusses the implications of these findings for closed-loop BMIs and outlines directions for future work,

including the incorporation of non-linear dynamics and adaptive re-calibration mechanisms.

II. METHODS

A. Experimental Setup and Data Acquisition

In this study, neural and kinematic data were acquired from a non-human primate (NHP) performing a centre-out reaching task towards eight distinct target directions. Neural signals were recorded from an electrode array implanted in the primary motor cortex, and spike trains were sampled at a resolution of $\Delta t = 1$ ms. Simultaneously, the hand position was tracked in Cartesian (x, y) coordinates, yielding a time series of two-dimensional spatial data. Each trial comprises an $N \times T$ spike matrix and a $2 \times T$ position trace, where N is the number of sorted units ($N = 98$) and T varies across trials. To facilitate batch processing, all trials were zero-padded to the global maximum duration $T_{\max}$. The validation of system performance was carried out with a task specific score, calculated from the Root Mean Squared Error (RMSE) between predicted hand trajectories and the ground-truth data, as well as the total running time of the system, encouraging solutions capable of operating within real-time constraints. The Task Score is calculated as $0.9 \times \text{RMSE} + 0.1 \times \text{Time}$.

B. Pre-processing

Neural spike train data underwent pre-processing to address inherent noise, dimensionality, and sparsity challenges, yielding a compact representation that enhanced decoder compatibility while preserving causality. This approach improved robustness for monkey arm trajectory decoding and facilitated cross-trial generalisation. We systematically explored three hyperparameter families: spike-count bin width, variance-stabilising transforms, and temporal smoothing. Model performance was evaluated on held-out validation data using a task-specific metric combining trajectory prediction accuracy (RMSE) and computational efficiency (refer to Section III).

a) Bin width and transform: Spike aggregation over time windows reduces noise by smoothing out millisecond-level fluctuations, which often arise due to environmental noise and are irrelevant for decoding purposes. The range of 5–25 ms was tested to balance temporal resolution (shorter bins preserve fine-grained information) and noise reduction (longer bins average out variability).

Next, several variance-stabilising transforms were systematically evaluated to address the Poisson-like characteristics of neural spike counts. The implemented transforms included square root, logarithmic, Anscombe, and power transformations ($x^{0.25}$), each designed to normalise the statistical distribution of spike counts for optimal linear modelling. The Anscombe transform, specifically engineered for Poisson-distributed data, provided superior performance by approximating a normal distribution with consistent variance. Empirical validation demonstrated that combining the Anscombe transform with 10 ms binning produced optimal decoding results as shown in Table II.

b) Temporal smoothing: Temporal smoothing was explored to further reduce noise in the binned spike counts by leveraging temporal correlations in neural firing, as arm movements are smooth and continuous. *Gaussian kernels* emphasise nearby time points, *moving averages* provide uniform smoothing, and *exponential kernels* prioritise recent spikes, aligning with the causal requirement of the task. These methods were tested to determine if additional smoothing could improve trajectory prediction by reducing high-frequency noise. However, the marginal RMSE improvement (0.04 cm) versus increased computation time (0.5–1 s) suggested that binning alone was sufficient for noise reduction, and smoothing added unnecessary complexity without significant performance gains. Disabling smoothing simplified the pipeline while maintaining performance.

c) Low-firing units: Low-firing unit removal was evaluated with thresholds from 0.1–1.0 Hz, based on the hypothesis that sparse firing reflects low signal-to-noise ratios and limited task relevance. However, testing revealed minimal accuracy improvements coupled with increased computation time, indicating that even sparsely firing neurons contain direction-specific movement information. We retained all 98 neural units, preserving the complete representation from both single- and multi-neuron recordings that collectively encode complementary aspects of arm trajectories.

C. Decoding Framework

The decoding framework is designed to predict both the intended reach direction and the continuous hand position. The framework is divided into two modules: a time-resolved *classification module* and a time-adaptive *regression module*. The former is responsible for predicting the reach angle based on the cumulative spike counts, whereas the latter estimates the hand position by mapping the neural activity to spatial coordinates.

Both modules operate on 10 ms, Anscombe-transformed spike bins (Section II: Pre-processing). The rationale, model families considered, and final design choices are outlined below; quantitative comparisons appear in Section III.

Angle Classification: A suite of multi-class classifiers was evaluated to convert cumulative spike counts into one of eight reach directions. Linear discriminant analysis (LDA) was adopted as the baseline because its shared-covariance assumption aligns with the approximately homoscedastic structure of the Anscombe-transformed counts and enables an efficient closed-form solution for computing discriminant scores. Support-vector machines with error-correcting output codes (SVM-ECOC) were included for their margin-maximising properties [4], and *k*-nearest neighbours (KNN) served as a non-parametric control able to exploit local structure after PCA [5]. To test the impact of more flexible decision boundaries we also trialled quadratic discriminant analysis (QDA) [6], decision trees [7], random forests (RF) [8] and naïve Bayes (NB) [9]. Finally, logistic regression with ECOC provided a linear probabilistic alternative to LDA [10].

LDA emerged as the empirically best classifier with a classification accuracy of 98.86%, the fastest run-time, and the best produced final task specific score. This intuitively is justified because its linear decision surfaces matched the approximately Gaussian distribution of Anscombe-stabilised counts and its pooled-covariance estimate mitigated over-fitting. A two-dimensional visualisation of the final feature space and its LDA decision regions is provided in Figure 1 (Section III).

Timing parameters: To balance early feedback against final accuracy we cross-validated the classifier window onset $t_{\text{start}} \in \{240, 280, 320\}$ ms and update interval $\Delta t_{\text{step}} \in \{10, 20, 40\}$ ms. The combination 320 ms / 20 ms gave the smallest Task Score without

significant loss of early-latency accuracy and is adopted throughout the paper. Note that the *binning window* (10 ms) concerns spike aggregation, whereas Δt_{step} determines how often the cumulative sum is re-evaluated; each classification update therefore integrates exactly two new bins. A three-update majority filter suppresses transient label flips.

Trajectory Regression: After classification, cumulative counts are centred, augmented with a bias and projected onto the first $D_{\text{PCA}} = 20$ principal components of a direction-specific basis. Five regressors were benchmarked:

- i. **Ordinary least squares (OLS)** – baseline, closed-form weights.
- ii. **Gaussian process regression (GPR)** – non-parametric, probabilistic; squared-exponential kernel.
- iii. **Support-vector regression (SVR)** – RBF kernel, ε -insensitive loss.
- iv. **LSBoost** – 100 shallow trees, learning rate 0.1.
- v. **XGBoost** – 200 trees, depth 3, learning rate 0.05.

OLS was retained for real-time decoding because it provided sub-centimetre accuracy with the lowest computation time, whereas GPR offered a marginal accuracy gain at more than twice the cost.

All regressors used the same training–testing splits as the classifiers. Hyper-parameters were kept at MATLAB defaults except for GPR noise variance (empirically set to 10^{-3}).

D. Implementation Details

All classifiers and regressors were implemented in MATLAB R2024a and executed on a single CPU core (3.4 GHz, 32GB RAM). We implement a 70/30 split of our training dataset for training/validation respectively. Principal components are recomputed for each direction/latency pair to account for the evolving covariance structure of cumulative counts. Table I lists the timing and dimensionality settings common to every experiment, while optimisation outcomes for binning and smoothing are presented later in Tables II–III.

TABLE I
FIXED IMPLEMENTATION PARAMETERS

Parameter	Value
Spike sampling resolution Δt	1 ms (raw)
Binning window ΔT_{bin}	10 ms (Anscombe)
Classifier window $[t_{\text{start}}, t_{\text{end}}]$	[320 ms, 560 ms]
Classifier update interval Δt_{step}	20 ms
Principal components D_{PCA}	20 (max)
Direction-switch filter length	3 updates

III. RESULTS

a) Preprocessing: Spike counts were aggregated in fixed windows of $\{5, 10, 15, 20, 25\}$ ms and subjected to one of five variance-stabilising transforms: square-root, logarithm, Anscombe, power-law ($x^{0.25}$), or none. Performance was evaluated with the *Task Score* = $0.9 \times \text{RMSE} + 0.1 \times \text{Time}$, which weights trajectory error (higher priority) against run-time. Table II shows that a 10 ms bin combined with the Anscombe transform yields the lowest score (9.65) as well as the minimum RMSE (8.22 cm). Both shorter (5 ms) and longer (15–25 ms) bins increased the score owing to either greater latency or reduced accuracy.

To test whether additional temporal filtering could improve the 20 ms/none baseline used in early regression experiments, Gaussian, moving-average and exponential kernels were applied to binned counts (Table III). The best smoothed configuration (moving average, 50 samples) reduced RMSE by only 0.04 cm yet increased the Task

Score from 11.08 to 11.09. Smoothing was therefore disabled in the final pipeline.

Removing units that fired below 0.1–1.0 Hz produced Task Scores in the range 11.0–11.3, offering no advantage; all 98 units were retained.

TABLE II
INFLUENCE OF BIN WIDTH AND VARIANCE-STABILISING TRANSFORM ON VALIDATION PERFORMANCE
Boldface indicates the setting used in the final pipeline

Bin width	Transform	RMSE (cm)	Time (s)	Task Score
25 ms	Anscombe	12.13	23.85	13.30
20 ms	none	9.93	20.21	10.96
15 ms	Anscombe	8.44	22.73	9.87
10 ms	Anscombe	8.22	22.53	9.65
5 ms	Anscombe	8.07	26.12	9.88

TABLE III
EFFECT OF TEMPORAL SMOOTHING ON DECODING ERROR
(20 MS BINS, NO TRANSFORM)

Kernel / parameter	RMSE (cm)	Time (s)	Task Score
None (baseline)	9.93	21.38	11.08
Gaussian $\sigma = 25$	10.03	25.31	11.56
Moving average 50	9.89	21.90	11.09
Exponential $\alpha = 0.3$	9.97	23.75	11.35

b) Angle-classifier comparison: Table IV reports cross-validated performance for the nine candidate models. LDA stands out with the lowest RMSE (9.93 cm) and the shortest training time (16.7 s), yielding the best Task Score of 10.61.

Logistic regression and SVM-ECOC achieve comparable label accuracies (98.5–98.9 %), but both post higher spatial error and longer run-times, demonstrating that classification accuracy alone is not a reliable proxy for kinematic precision. This discrepancy arises because occasional “long-jump” mistakes - e.g. confusing a target with its opposite - contribute disproportionately to Euclidean error, whereas the rare misclassifications made by LDA fall almost exclusively on neighbouring angles and therefore incur smaller penalties.

Non-parametric or high-capacity models (KNN, QDA, decision tree, random forest) illustrate the opposite trade-off: faster training or greater flexibility does not offset their tendency to over-fit, resulting in RMSEs well above 13 cm.

The geometric basis for LDA’s lead is evident in Figure 1: after variance stabilisation and PCA, the eight reach directions form compact clusters that are almost perfectly separated by linear boundaries; the few misclassified points lie on cluster borders, matching the pattern seen in Table IV.

TABLE IV
CLASSIFIER BENCHMARKING (10MS BINS, ANSCOMBE TRANSFORM)

Classifier	RMSE (cm)	Time (s)	Task Score	Accuracy (%)
LDA	9.93	16.74	10.61	98.86
SVM-ECOC	10.99	20.27	11.92	98.41
KNN	13.42	8.59	12.94	96.22
QDA	27.08	10.76	25.44	76.94
Decision tree	28.68	13.42	27.15	80.30
Random forest	14.79	105.75	23.89	95.35
Logistic reg. (ECOC)	10.52	30.52	12.52	98.88
Naïve Bayes	14.31	470.51	59.93	95.60

c) Trajectory-regressor comparison: Direction-conditioned regressors are compared in Table V. GPR produced the best RMSE

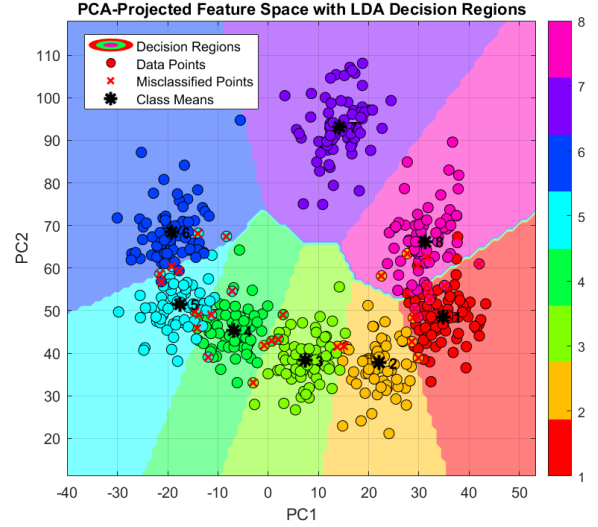


Fig. 1. Two-dimensional PCA projection of cumulative spike counts at $t = 560$ ms. Colours indicate LDA decision regions for the eight reach directions; filled circles are test trials, red crosses mark the few misclassified points, and black stars denote class means. The first two principal components explain 42.8 % of the variance, confirming that angles are almost linearly separable after variance stabilisation.

(7.98 cm), but took more than twice as long to train as OLS. OLS balanced accuracy (9.95 cm) with the fastest run-time (10.6 s), achieving the best overall Task Score (10.01) and was therefore selected for the online pipeline.

Figure 2 shows how the OLS decoder tracks a typical reach. Predicted points stay close to the ground-truth path from the very first update, and the colour gradient confirms that this tight alignment is maintained throughout the 240 ms prediction window. Trajectories remain smooth, with no consistent overshoot or drift at the target, indicating that the simple linear mapping in PCA space captures the dominant hand dynamics for this task.

TABLE V
REGRESSOR BENCHMARKING (DIRECTION-SPECIFIC, 20 PCs).

Regressor	RMSE (cm)	Time (s)	Task Score
Gaussian Process	7.98	25.54	9.73
Support-vector	8.54	20.62	9.75
OLS (Linear)	8.22	22.54	9.65
LSBoost	9.60	133.33	21.97
XGBoost	9.60	131.54	21.79
Bagged trees	8.10	144.31	21.72

IV. DISCUSSION

Decoding arm kinematics from cortical spiking activity is fundamentally a signal-to-noise problem: information about a smooth, low-dimensional limb trajectory is hidden in high-dimensional, stochastic point processes. Our study shows that much of this noise can be neutralised with a minimally-complex architecture. Binning spikes at 10 ms followed by an Anscombe variance stabiliser standardises the count distribution and, crucially, delivers features that are both Gaussian-like and linearly separable. More aggressive interventions - kernel smoothing or channel pruning - offered only marginal gains once counts were stabilised, suggesting that the motor-cortex population already contains a robust redundant code when sampled at millisecond resolution.

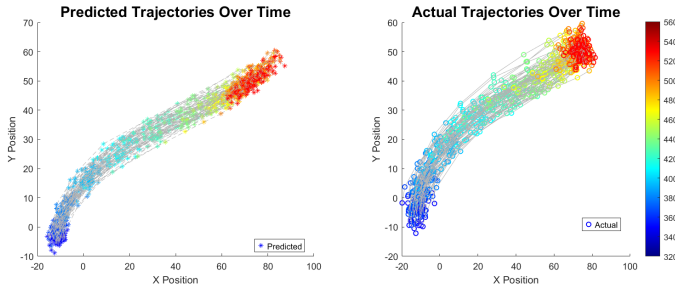


Fig. 2. OLS-predicted (left) and Actual (right) hand positions for all test trials of one reach direction. Colour encodes time from 320 ms (dark blue) to 560 ms (dark red); grey lines connect successive points.

On this cleaned feature space a simple linear discriminant was able to outperform more elaborate classifiers. The success of LDA is less surprising in retrospect: by pooling covariance across classes it allocates parameters where the data support them, while its closed-form decision rule makes it immune to the kernel-width and regularisation pitfalls that often hamper SVMs or KNNs. Just as importantly, decoding performance saturates after roughly 560 ms of integration, which can be explained by the hand movement stopping. Our model ignores data beyond 560ms (to reduce complexity). In a real-life deployment, this would have to be made dynamic; the model should be able to decode the end of the hand-motion without hard-coding. Regardless, further complexity adds computational latency with little benefit. The three-update majority filter then acts as a lightweight temporal prior, dampening the occasional outlier without the inertia of a full Kalman smoother.

Continuous decoding presents a more delicate trade-off. Gaussian-process regression squeezes the last centimetre of error from the data, but its cubic training cost places it beyond the real-time budget of a tethered experimental rig, let alone an implanted system. Ordinary least squares, when preceded by a modest PCA, is sufficient: by rotating the feature basis it captures the dominant firing patterns associated with each reach direction, leaving a diagonal covariance structure that a linear regressor can model efficiently. The resulting latency - on the order of tens of milliseconds - comfortably meets the 100 ms control-loop requirement of contemporary intracortical prostheses.

Two features of the population code deserve emphasis. First, information was surprisingly resilient to electrode dropout; removing one fifth of the channels inflated error by a third of a centimetre. This redundancy is encouraging for chronic implants, where channel loss is inevitable. Second, performance plateaued once ten principal components were retained, implying that reach-related dynamics are confined to a very low-dimensional manifold - a result consistent with recent dynamical systems accounts of motor cortex [11]. Future work could exploit this structure more directly, for instance by fitting latent-state models whose dynamics evolve on that manifold rather than in the original spike space.

The present study remains open-loop, so we cannot yet claim closed-loop robustness. Feedback will almost certainly reshape firing statistics, although the simplicity of LDA and OLS should make online adaptation straightforward - covariance and weight matrices can be updated with rank-one formulas without retraining from scratch. Extending the decoder to three-dimensional reaches and velocity would be a natural next step, as would testing whether a sparsified Gaussian-process kernel can bring sub-centimetre accuracy into the real-time domain.

Despite the performance achieved by our decoder, certain characteristics of the task itself raise questions regarding translation to realistic engineering or clinical applications - the task is at risk of overfitting to artificial features of the dataset that lack translational relevance. Firstly, in realistic BMI applications, movements do not occur in predetermined segmented trials with fixed onset times; here, the task allows the system to implicitly use movement stereotypes which would fail under online scenarios (despite performing well in silico). Secondly, and perhaps more importantly, the movements are constrained to a fixed set of directions with consistent kinematic profiles (enabling the discrete movement-class decoder, and the state-based trajectory estimator); here the system learns these implicit templates rather than truly decoding movement intention (as seen in Figure 2, where the decoded trajectories have more uniform kinematics than the true trajectories). While this leads to a high task score, it risks poor generalisation to real-life dynamic movements.

V. CONCLUSION

We have demonstrated that a two-stage decoder built from well-understood linear techniques can match, and in some cases surpass, the accuracy of heavier non-linear models while staying comfortably within real-time constraints. A 10 ms / Anscombe pre-processing front-end feeds an LDA classifier updated every 20 ms, whose output selects a direction-specific PCA + OLS regressor. The architecture is computationally lightweight, tolerant to channel loss and achieves near-saturation accuracy within the first half-second of movement. These qualities make it an attractive candidate for closed-loop brain-machine interfaces, where reliability and speed often outweigh marginal gains in endpoint precision. Ongoing work will focus on closed-loop validation, extension to richer motor tasks and incremental updates that allow the decoder to track long-term neural non-stationarities without full retraining.

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